General Science

Alveary

About the Course

This is the first of a two-year Physical Science course (i.e., introductory physics and chemistry). The fullness of the course includes topics in earth science, history, politics, and more. The completion of this course provides a gentle yet robust transition into the high school disciplines. The lab component is required for this course. Coordinating afternoon activities are provided in Outdoor Work.

This topic is included in the following course(s): Science: Grade 7

About Science: Grade 7
In level 7, learners extend their relationship with science in a new direction (time), situating science in its historical, political, and cultural context, and beginning to understand that science is an ever-changing process rather than a static body of knowledge. Reading level and expectation increase from previous levels.

Science: Grade 7
The Big Picture:
To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. In Form 1, students became acquainted with this area of knowledge, science. Form 2 students extended their scope and began to encounter science as an active part of society. Form 3 students extend their relationship with science in a new direction (time) and consider science as a process by which man seeks Truth within Creation. They begin to situate science in its historical, political, and cultural context with all of the complexities of man's own story. In this way, students are prepared to think about science with a complete and holistic perspective.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just "feels right." Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Citizenship Grades 4+).

Placement & Combining Tips

About Science: Grade 7
In level 7, learners extend their relationship with science in a new direction (time), situating science in its historical, political, and cultural context, and begin to understand that science is an ever-changing process rather than a static body of knowledge. Reading level and expectations increase from previous levels.

General Science: Grade 7
For seventh-grade students or possibly hungry sixth-graders or eighth-graders taking their time. If combining grades is desirable: the combined grades may begin at the appropriate level and move forward together, or teachers can adapt the reading level and laboratory content to meet their needs.

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
7	General Science: Grade 7 2 times/week 30 min	The Story of Science: Aristotle Leads the Way

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Grade 7				
Natural History: Grade 7	General Science: Grade 7	General Science: Grade 7	Nature Notebook: Grade 7	Labs: Grade 7 Nature Walks & Scouting: Grades 1-8

Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 7
☐ Obtain any supplies indicated on the science or grade-level supply lists.
☐ Download any apps and shortcut any desired links.

Nature Lore:
☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings, except where the suggestion is to collect some Thing for a lesson.
☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Special Topics & Field Trips

Science: Grade 7
Encourage students to choose their own special topic and to notice its ecological relationships. What or who are they curious about or interested in getting to know better? Teachers can choose their own special study, too!

☐ Learn a few of your local species or varieties in connection with special topics.

General Science: Grade 7
☐ Terms 1 and 3: a planetarium or local astronomy club event
☐ Term 2: a playground

Term Prep & Teacher Tips

General Science: Grade 7
☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:
• medium-large sized recycled box, roughly 1-2'
• knife, blade, or scissors to cut the box
• table/reading lamp

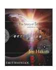
• table or countertop
• books or wood blocks to adjust height
• 1 gallon jug of milk or water, the type with a handle
• horizontal bar to support weight, such as broom handle held by 2 people. a chin up bar, a porch railing, etc.
• recycled cereal box or toothbrush
• a flat location that is in the sun at noon
• various supplies by student design
• printables
• computer
• optional: empty cereal box
• optional: orange
• optional: flashlight
• optional: calculator, such as found on most electronic devices

Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ View Book List Details

General Science: Grade 7



The Story of Science: Aristotle Leads the Way

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ View Basic Supplies
∞ View Supply List Details

General Science: Grade 7

Household Items - General Science: Grade 7

Quick Links

Click THIS text or scan the QR code for links.

(No Quick Links Assigned)

General Science: Grade 7

How To Teach

Introduce

• Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap.

• Often, the introduction in the lesson plans or the title of the book's section can help learners to draw out the main idea. Use this to support learners, as appropriate.

• Some learners may benefit from using pictures or looking back briefly.

• Allow them time to share any concerns and come alongside, as needed.

Read

• Read or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their journal, notebook, or other paper available in case they need to draw or diagram during the lesson.

• Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.

• If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, and beyond.

Narrate

• Process the ideas of the lesson by retelling events, describing, explaining a concept, etc. Tips about helping learners deal with various types of passages are provided, but teachers can learn more in Charlotte Mason's School Education Chapter 16, especially p.180.

• Learners may use words, pictures, Legos, PowerPoint, etc, to process and convey ideas in their own way.

• Teachers may take turns to model.

Discuss

• Consider together any thoughts, confusion, or concerns about the passage, keeping in mind that oftentimes new concepts are not going to be 'mastered' on the first or even second introduction. Mastery is not necessarily the goal, but curiosity and thinking.

• Questions/topics for further discussion are often provided in the lesson plans (or even lab books) to help. There are no right or wrong answers to these. Alternatively, many of these can be used for composition, depending on the needs of the learner and the instructional goals of the teacher. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.

• Note that learners may need to spend more than the allotted time engaging with or even repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.

• Notice if there were any dates that they want to keep for their Book of Centuries.

Connect

• Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner needs and interest.

• These can be viewed as alternative ways to engage.

General Science: Grade 7

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Term 1

WEEK 1

⬚ 30m General Science: Grade 7 - Lesson 1

Thinking About Creation

▯ Materials: Aristotle Leads the Way

PREP: Read Teacher Tip

→ INTRO
This book tells the story of how modern science developed, beginning with the ideas of the ancient Greeks. Most of them wondered and learned from the heavens and from mathematics, eventually defining science. There are many interesting sidebars and tangents in the text, but read the main text first and then explore them, as desired. Watch the course introduction video:
∞ Video Link: Form 3 Science Intro (5:04)

→ READ, NARRATE, & DISCUSS
Ch.1 p.1-8 "The universe" - "to have them."
• Tell about the earliest beginnings of science in the ancient world. How did men begin?
• The author of this book invites all learners of different beliefs to find common ground in this discussion about how we understand the world we share. Discuss some of the ways she does this.

→ SUPPLEMENTAL
∞ Video Link: NatGeo Mesopotamia (4:10)
∞ Video Link: Mayan Creation Story (2:55)
∞ Video Link: Hindu Creation Story (2:41)

★ TEACHER TIP
Watch the course introduction video with your student(s).

★ TEACHER TIP
Even as students begin to work more independently, teacher engagement is very important. Discussion prompts are provided to help, if natural discussion is a challenge.

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

• IMPORTANT DATES
Sumerian Empire (c.3100 B.C.-c.2000 B.C.), Egyptian Civilization (c.3100 B.C.-332 B.C.)

WEEK 1

⬚ 30m General Science: Grade 7 - Lesson 2

What is a Myth?

▯ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.2 p.9-14 "Once, or so" - "makes us human."
• According to the author, what is the distinction between myth and science? And what is NOT the distinction? Is there anything to add based on your own tradition?
• Why is myth important? If you have read much of J.R.R.Tolkien's work, consider the role myth plays in his stories. Compare/contrast both authors' ideas about myth. If you haven't yet read his work, consider another author who makes use of myth, or file away this idea for the day that you do read his work!
• What is a hypothesis?
• How is science (and its limits) part of what makes us human?

→ SUPPLEMENTAL
∞ Video Link: Finding Summer Constellations (5:39)

★ TEACHER NOTE
The author invites readers to consider what is myth and what is science, a consideration that engages with religion, on p.11-12: "How do we" - "fiction usually does." Teachers might choose to allow the idea to pass by the way or to engage in discussion.

WEEK 2

⬚ 30m General Science: Grade 7 - Lesson 3

What is a Myth?

▯ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.2 p.14-19 "We've learned" - "to be born."
• Describe the process of science.
• "The world is God's epistle written to mankind... It was written in mathematical letters." Discuss.
• The author says that the "marriage of numbers and nature... is an essential part of the scientific story." Why? Do you agree? Where have you noticed numbers in nature?
• Why do you suppose the story begins with sky watching?

→ SUPPLEMENTAL
∞ Video Link: Is Math Invented or Discovered? (5:11)

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 2

⬚ 30m General Science: Grade 7 - Lesson 4

Thinking about Time

▯ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

• IMPORTANT DATES
Egyptian Solar Calendar (c.4th century B.C.)

General Science: Grade 7

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Term 1

→ **READ, NARRATE, & DISCUSS**
Ch.3 p.20-25 "Perhaps it was" - "around the globe."
• What kinds of observations and what kinds of questions did early sky watchers have?
• Explain the similarities and differences between the lunar calendar and the solar calendar. Which is closer to the calendar we use?

→ **SUPPLEMENTAL**
∞ Video Link: The Analemma (4:23)
∞ Video Link: Want to measure the analemma? (7:33)

WEEK 3

⬚ **30m General Science: Grade 7 - Lesson 5**
Phases of the Moon

Materials: Aristotle Leads the Way

→ **RECAP**
What did you read last time?

→ **READ, NARRATE, & DISCUSS**
Ch.3 p.26-27 "For a long time," - "facing the sun."
• Explain the moon's phases.

→ **SUPPLEMENTAL**
∞ Video Link: Moon Phases (9:45)

• **NATURE NOTEBOOK**
Prompt for General Science in Outdoor Work Quick Link.

WEEK 3

⬚ **30m General Science: Grade 7 - Lesson 6**
Thinking about Time

Materials: Aristotle Leads the Way

→ **RECAP**
What did you read last time?

→ **READ, NARRATE, & DISCUSS**
Ch.3 p.28-31 "Palenque," - "our solar system."
• Were the ancient calendar makers doing science? Why or why not? If not science, what was it?
• What ideas have been passed down to us from the ancient calendar makers?

→ **SUPPLEMENTAL**
∞ Video Link: Decoding the Astronomy of Stonehenge (6:29)

• **FIELD TRIP**
Visit a planetarium or astronomy club event.

WEEK 4

⬚ **30m General Science: Grade 7 - Lesson 7**
Ancient Record Keeping

Materials: Aristotle Leads the Way

→ **RECAP**
What did you read last time?

→ **READ, NARRATE, & DISCUSS**
Ch.3 p.32-33 "When did people" - "closely guarded secret."
• Discuss the beginning of math.

→ **SUPPLEMENTAL**
∞ Video Link: Ancient Writing and Modern Tech (6:44)

• **NATURE NOTEBOOK**
Prompt for General Science in Outdoor Work Quick Link.

WEEK 4

⬚ **30m General Science: Grade 7 - Lesson 8**
Thinking about Nature

Materials: Aristotle Leads the Way

→ **RECAP**
What did you read last time?

→ **READ, NARRATE, & DISCUSS**
Ch.4 p.34-38 "Does where we live" - "scientific approach."
• Set the scene for science in Ionia.
• How was the observation of patterns in nature an important concept?

• **IMPORTANT DATES**
Thales (c.636 B.C.-546 B.C.)

WEEK 5

⬚ **30m General Science: Grade 7 - Lesson 9**
Thinking about Nature

Materials: Aristotle Leads the Way

→ **RECAP**
What did you read last time?

→ **READ, NARRATE, & DISCUSS**

• **NATURE NOTEBOOK**
Prompt for General Science in Outdoor Work Quick Link.

General Science: Grade 7

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Term 1

Ch.4 p.38-41 "We don't know" - "is heating up."
• Why was Thales important?
• A recurring idea: how is an incorrect idea still important?
• "While others trembled at nature's wonders and puzzles, the Ionians asked hard questions." Discuss.

➔ SUPPLEMENTAL
∞ Video Link: Law vs. Theory (5:12)

WEEK 5

⬚ 30m General Science: Grade 7 - Lesson 10

Solving Real-Life Problems with Math

▯ Materials: Aristotle Leads the Way

PREP: Read Teacher Tip

➔ RECAP
What did you read last time?

➔ READ, NARRATE, & DISCUSS
Ch.4 p.42-43 "The eye can" - "about 350 paces."
• Explain how one can measure from a distance.

★ TEACHER TIP
Are your learner(s) demonstrating more familiarity/interest in the habits of observation, measurement, and/or record-keeping? More interest/ease in discussing the challenges of early scientists in their social context? If not, consider observing and discussing together more often, exploring extra helpings, or taking a field trip to a planetarium. Reach out through Contact Us, if you need help.

WEEK 6

⬚ 30m General Science: Grade 7 - Lesson 11

Dangerous Ideas

▯ Materials: Aristotle Leads the Way

PREP: Read Teacher Tip

➔ RECAP
What did you read last time?

➔ READ, NARRATE, & DISCUSS
Ch.5 p.44-49 "Thales did" - "started so well."
• The author states that scientific questioning "was threatening to the keepers of the old ideas." Why do people sometimes feel threatened by new ways of thinking?
• What kinds of ideas did the scientists in this chapter think about?
• What was the danger of Anaxagoras' ideas? Can science be dangerous? If so, under what circumstances?

★ TEACHER TIP
Remember to engage students on their reading through natural discussion. If you need help, use the bulleted prompts.

★ TEACHER NOTE
Evolution is mentioned in the third paragraph on p.44. Teachers might choose to allow it to pass by the way or to discuss with students.

• IMPORTANT DATES
Anaximander (c.611 B.C.-c.547 B.C.), Anaximenes (c.570-500 B.C.), Anaxagoras (c.500-428 B.C.)

WEEK 6

⬚ 30m General Science: Grade 7 - Lesson 12

The Origins of Math

▯ Materials: Aristotle Leads the Way

➔ RECAP
What did you read last time?

➔ READ, NARRATE, & DISCUSS
Ch.5 p.50-53 "It was the" - "in Egyptian tombs."
• Discuss how the concept of zero changed mathematics.

WEEK 7

⬚ 30m General Science: Grade 7 - Lesson 13

Elements

▯ Materials: Aristotle Leads the Way

PREP: Read Teacher Tip

➔ RECAP
What did you read last time?

➔ READ, NARRATE, & DISCUSS
Ch.6 p.54-57 "Thales said" - "a small country."
• What's the 'big idea' or 'main purpose' of this chapter? Use the title to help, if it's not immediately clear.
• How does this chapter connect back to ideas from previous chapters? At the same time, how does it move the story forward?

★ TEACHER TIP
Help students begin to discern the main ideas of a chapter and link this to previous information.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

• IMPORTANT DATES
Empedocles (c.495 B.C.-c.434 B.C.)

WEEK 7

⬚ 30m General Science: Grade 7 - Lesson 14

Thinking at Sea

▯ Materials: Aristotle Leads the Way

➔ RECAP
What did you read last time?

General Science: Grade 7

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Term 1

→ READ, NARRATE, & DISCUSS
Ch.7 p.58-61 "Herodotus" - "of tall tales."
• Tell about Herodotus.
• How did sea exploration add to scientific observation?
• Discuss how the sailors and Herodotus were both engaging in scientific thinking.

WEEK 8

⬚ 30m General Science: Grade 7 - Lesson 15
Polestars

▯ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.7 p.62-63 "The North Star" - "show the way."
• Tell what you know about pole stars.

★ TEACHER NOTE
The timescale of Earth's existence is mentioned on p.63: "takes about 25,800 years." Teachers might choose to allow it to pass by the way or to discuss alternative theories.

WEEK 8

⬚ 30m General Science: Grade 7 - Lesson 16
Science and Math

▯ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.8 p.64-69 "A Sense of Number" - "change the world."
• Tell the story of the birth of Pythagoras.
• Discuss the importance of math.
• Try solving magic squares. The magic numbers for some squares are as follows: 4x4 is 34, 6x6 is 111, 7x7 is 175, 9x9 is 369. Can you find the number for a 5x5 square?

• IMPORTANT DATES
Pythagoras (c.582 B.C.-507 B.C.)

WEEK 9

⬚ 30m General Science: Grade 7 - Lesson 17
Pi

▯ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.8 p.70-71 "Of all the" - "the ancient world."
• What is pi and why is it elusive?
• In the video, note what a challenge it can be to find the source of bias in an experiment and eliminate it! Try playing with pi. Knowing that circumference equals pi times diameter, find various objects for which you can measure either circumference or diameter, and then calculate the other. If the circumference of the Earth is 24,901 miles, what is the diameter?

→ SUPPLEMENTAL
∞ Video Link: Finding Pi with Darts (5:48)

WEEK 9

⬚ 30m General Science: Grade 7 - Lesson 18
Pythagoras

▯ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.9 p.72-75 "Samos was part" - "if it really happened."
• What kinds of data have these early scientists been collecting so far? What do the Greeks bring? Is anything still wanting?
• Pythagoras "thought numbers were... an expression of God's mind." Discuss. Is mathematics a language? If so, does God use this language?

WEEK 10

⬚ 30m General Science: Grade 7 - Lesson 19
Pythagoras

▯ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.9 p.75-81 "Remember, the" - "has done more."
• Were Pythagoras' ideas good? Were they dangerous? Why or why not? Good or dangerous, were

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

Term 1

they important?
• How does this chapter connect back to ideas from previous chapters? At the same time, how does it move the story forward?
• Notice the shapes toward the bottom of the article and try drawing your own fractals from any starting line, curve, or shape!

→ SUPPLEMENTAL
∞ Video Link: How to Prove the Pythagorean Theorem (5:17)
∞ Article Link: Fractals in Nature

WEEK 10

⬚ 30m General Science: Grade 7 - Lesson 20
Thinking about Irrational Numbers

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.9 p.82-85 "The Pythagoreans" - "in a pentagram."
• Draw a picture that shows or explains the importance of an irrational number.

→ SUPPLEMENTAL
∞ Video Link: Making Sense of Irrational Numbers (4:41)

WEEK 11

⬚ 30m General Science: Grade 7 - Lesson 21
Thinking about Atoms

Materials: Aristotle Leads the Way

PREP: Read Teacher Tip

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.10 p.86-91 "I would rather" - "believed in atoms."
• Discuss how Democritus took the idea of elements further.
• Explain the central idea of Democritus' atom.
• Discuss how science moves beyond an initial idea and explain why the ancient Greeks could not move forward.

★ TEACHER TIP
When student(s) take exams next week, evaluate what they recall, but more importantly whether the practical habits of science or the challenges of early scientists are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc? Reach out through Contact Us, if you need help.

• IMPORTANT DATES
Democritus (c.460 B.C.-c.370 B.C.), Socrates (469-399 B.C.), Aristotle (384-322 B.C.)

WEEK 11

⬚ 30m General Science: Grade 7 - Lesson 22
A Poem on the Nature of Things

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.10 p.92-93 "Lucretius" - "one oxygen atom."
• What kinds of observations does Lucretius note in his poem?

WEEK 12

⬚ 30m General Science: Grade 7 - Lesson 23
Exam Week

→ EXAMS
Answer question(s) related to the course.

Questions will come from:
Aristotle Leads the Way

WEEK 12

⬚ 30m General Science: Grade 7 - Lesson 24
Exam Week

→ EXAMS
Answer question(s) related to the course.

Questions will come from:
Aristotle Leads the Way

General Science: Grade 7

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Term 2

WEEK 1

⬚ **30m General Science: Grade 7 - Lesson 25**
Aristotle

▯ Materials: Aristotle Leads the Way

PREP: Read Teacher Tip

→ RECAP
Where did the story leave off? Look back if you need help.

→ READ, NARRATE, & DISCUSS
Ch.11 p.94-97 "By the time" - "those ideas well."
• What's the 'big idea' or 'main purpose' of this chapter? Use the title to help, if it's not immediately clear.
• Explain Plato's philosophy.

★ TEACHER TIP
Remember to engage students on their reading through natural discussion. If you need help, use the bulleted prompts.

• IMPORTANT DATES
Plato (c.427 B.C.-347 B.C.)

WEEK 1

⬚ **30m General Science: Grade 7 - Lesson 26**
Plato's Search for Perfection

▯ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.11 p.98-99 "Plato looked" - "in understanding them."
• Try using cardboard or cardstock to make the 5 Platonic solids. How many of Archimedes' solids can you render or imagine?

WEEK 2

⬚ **30m General Science: Grade 7 - Lesson 27**
Aristotle

▯ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.11 p.100-105 "The discussed" - "good thinkers do."
• Compare/contrast Plato and Aristotle.
• How does this chapter connect back to ideas from previous chapters? At the same time, how does it move the story forward?

WEEK 2

⬚ **30m General Science: Grade 7 - Lesson 28**
Aristotle's Incorrect-Yet-Important Model of Earth

▯ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.12 p.106-111 "Funny thing" - "what they had."
• Notice again the theme that even incorrect ideas can be important. Discuss.
• Was Aristotle a genius who moved science forward, or was he responsible for holding science back?

WEEK 3

⬚ **30m General Science: Grade 7 - Lesson 29**
Wandering in the Sky

▯ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.12 p.112-113 "If you happen" - "merry-go-round."
• Explain why planetary movement is confusing to a sky watcher on Earth.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 3

⬚ **30m General Science: Grade 7 - Lesson 30**
A Very Different Model of Earth

▯ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS

• IMPORTANT DATES
Aristarchus (c.310 B.C.-c.230 B.C.)

General Science: Grade 7

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Term 2

Ch.13 p.114-117 "Aristotle's universe" - "their big ideas."
• What's the 'big idea' or 'main purpose' of this chapter?
• How does this chapter connect back to ideas from previous chapters? At the same time, how does it move the story forward?

WEEK 4

⬚ 30m General Science: Grade 7 - Lesson 31

How Far the Moon?

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.13 p.118-119 "How do you" - "came around."
• How do you use time to measure distance (without a laser)?

→ SUPPLEMENTAL
∞ Video Link: How Levers Work (4:46)

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 4

⬚ 30m General Science: Grade 7 - Lesson 32

The City of Alexandria

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.14 p.120-125 "The Mediterranean" - "astonishing city."
• Tell about Alexandria. Consider how Alexandria connects back to previous chapters and why it might be important moving forward.

• IMPORTANT DATES
Ptolemaic Dynasty (332 B.C-30 A.D.), City of Alexandria (est. 331 B.C.)

WEEK 5

⬚ 30m General Science: Grade 7 - Lesson 33

The Pharos Lighthouse

Materials: Aristotle Leads the Way

PREP: Read Teacher Tip

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.14 p.126-127 "A gigantic" - "Mediterranean Sea."
• Tell about the Pharos lighthouse.

★ TEACHER TIP
Are your learner(s) demonstrating more familiarity/interest in engineering or how Things work? More interest/ease in discussing the complex relationship between science and society? If not, consider observing and discussing together more often, exploring extra helpings, or taking a field trip to a play ground or science museum with an engineering exhibit. Reach out through Contact Us, if you need help.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 5

⬚ 30m General Science: Grade 7 - Lesson 34

Technology in Alexandria

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.15 p.128-133 "For centuries" - "to jest with."
• Describe technology at this time.
• Tell about Hero.

→ SUPPLEMENTAL
∞ Video Link: Human Powered Helicopter (5:36)

• FIELD TRIP
Visit a playground and notice the simple machines with which the equipment is designed.

• IMPORTANT DATES
Hero (1st century A.D.)

WEEK 6

⬚ 30m General Science: Grade 7 - Lesson 35

Hero's Air-Powered Machines

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.15 p.134-135 "The title" - "self-trimming wick."
• What's a pneumatic device?

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

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Term 2

WEEK 6

30m General Science: Grade 7 - Lesson 36
Euclid, Philosopher of Math

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.16 p.136-142 "A math professor" - "what I mean."
• Explain how Euclid took math "further" than business dealings and measurement.

• IMPORTANT DATES
Euclid (c.325 B.C.-c.270 B.C.)

WEEK 7

30m General Science: Grade 7 - Lesson 37
Thinking about Prime Numbers

Materials: Aristotle Leads the Way

PREP: Read Teacher Tip

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.16 p.143-145 "Do you like" - "of as 'molecules.'"
• Explain prime numbers.
• Try Eratosthenes' Sieve as described on page 145. How many primes can you find?

★ TEACHER TIP
Remember to engage students on their reading through natural discussion. If you need help, use the bulleted prompts.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 7

30m General Science: Grade 7 - Lesson 38
Engineering to Solve Problems

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.17 p.146-152 "It sounded" - "I have it!"
• If Hero was an engineer and Euclid was a mathematician, what was Archimedes?
• Archimedes seemed to have some mixed feelings about engineering. Explain. How do you think the ancients would feel about the trend toward biomimicry in our modern era?

→ SUPPLEMENTAL
Article Resource: Fascinating World of Biomimicry (5:59)

★ TEACHER NOTE
The age of the Earth is mentioned in the supplemental video (2:50 and 5:12). Teachers might choose to allow it to pass by the way, to discuss with students, or to skip this one.

• IMPORTANT DATES
Archimedes (c.287 B.C.-212 B.C.)

WEEK 8

30m General Science: Grade 7 - Lesson 39
Engineering to Solve Problems

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.17 p.152-157 "When Archimedes" - "of his brain."
• Tell the story of "Eureka!"
• Archimedes solved problems by looking at them differently. Discuss.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 8

30m General Science: Grade 7 - Lesson 40
How did the Claw Really Work?

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.17 p.158-159 "Did Archimedes" - "to find it."
• Why is the claw questioned when there are historical accounts of it?

WEEK 9

30m General Science: Grade 7 - Lesson 41
Measuring the Earth

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

• IMPORTANT DATES
Eratosthenes (c.275 B.C.-c.195 B.C.)

Term 2

➔ READ, NARRATE, & DISCUSS
Ch.18 p.160-163 "Eratosthenes" - "all about it."
• What's the 'big idea' or 'main purpose' of this chapter?
• How does this chapter connect back to ideas from previous chapters? At the same time, how does it add to the story?

➔ SUPPLEMENTAL
∞ Video Link: Six Simple Machines in One (7:08)

WEEK 9

⬚ 30m General Science: Grade 7 - Lesson 42
How did Eratosthenes Do It?

Materials: Aristotle Leads the Way

➔ RECAP
What did you read last time?

➔ READ, NARRATE, & DISCUSS
Ch.18 p.164-165 "Syne, near modern" - "24,855 miles."
• Explain how Eratosthenes found the circumference of the Earth.

WEEK 10

⬚ 30m General Science: Grade 7 - Lesson 43
The Roman Empire

Materials: Aristotle Leads the Way

➔ RECAP
What did you read last time?

➔ READ, NARRATE, & DISCUSS
Ch.19 p.166-169 "It was the" - "to be controlled."
• Discuss some positive and negative aspects of the shift from Greek to Roman prominence.
• How might the work of the Greeks from the last few chapters have been different in Roman culture?

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

• IMPORTANT DATES
Library of Alexandria burned (48 B.C.)

WEEK 10

⬚ 30m General Science: Grade 7 - Lesson 44
The Roman Empire

Materials: Aristotle Leads the Way

➔ RECAP
What did you read last time?

➔ READ, NARRATE, & DISCUSS
Ch.19 p.169-173 "Jerusalem was" - "Brueghel the Elder."
• Tell about Strabo.

• IMPORTANT DATES
Strabo (c.64 B.C.-c.24 A.D.)

WEEK 11

⬚ 30m General Science: Grade 7 - Lesson 45
Mapping Sky and Land

Materials: Aristotle Leads the Way

PREP: Read Teacher Tip

➔ RECAP
What did you read last time?

➔ READ, NARRATE, & DISCUSS
Ch.20 p.174-180 "It's easy" - "called triangulation."
• What's the 'big idea' or 'main purpose' of this chapter?
• How do the ideas connect back to previous chapters? At the same time, how does it move the story forward? What does the science seem to be building toward?

➔ SUPPLEMENTAL
∞ Video Link: Precession of Earth (3:13)

★ TEACHER TIP
When student(s) take exams next week, evaluate what they recall, but more importantly whether engineering, design, or the relationship between science and society are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc? Reach out through Contact Us, if you need help.

• IMPORTANT DATES
Hipparchus (c.190 B.C.-c.120 B.C.)

WEEK 11

⬚ 30m General Science: Grade 7 - Lesson 46
Points of Reference

Materials: Aristotle Leads the Way

➔ RECAP
What did you read last time?

➔ READ, NARRATE, & DISCUSS
Ch.20 p.181-183 "A point" - "(see page 207)."
• Describe how to chart stars.

• NATURE NOTEBOOK
Prompt for General Science in Quick Link.

General Science: Grade 7

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Term 2

WEEK 12	⬚ **30m General Science: Grade 7 - Lesson 47** *Exam Week* **→ EXAMS** Answer question(s) related to the course. Questions will come from: Aristotle Leads the Way	
WEEK 12	⬚ **30m General Science: Grade 7 - Lesson 48** *Exam Week* **→ EXAMS** Answer question(s) related to the course. Questions will come from: Aristotle Leads the Way	

General Science: Grade 7

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Term 3

WEEK 1

⬚ 30m General Science: Grade 7 - Lesson 49

Ptolemy the Thinker

▯ Materials: Aristotle Leads the Way

PREP: Read Teacher Tip

→ NOTE
The last three lessons of the year have been left open to complete unfinished readings, labs, projects, or science notebooks.

→ RECAP
Recall where we left the story. Look back if you need help.

→ READ, NARRATE, & DISCUSS
Ch.21 p.184-189 "Claudius Ptolemy" - "out of style."
• A lot of Ptolemy's ideas were incorrect. Why is he remembered as a great thinker?

→ SUPPLEMENTAL
∞ Video Link: Armillary Sphere (3:42)
∞ Video Link: Animation of How the Sphere Works (4:14)

★ TEACHER TIP
Remember to engage students on their reading through natural discussion. If you need help, use the bulleted prompts.

• IMPORTANT DATES
Claudius Ptolemy (c.85 A.D.-c. 165 A.D.)

WEEK 1

⬚ 30m General Science: Grade 7 - Lesson 50

The Material Versus the Spiritual

▯ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.22 p.190-193 "After some" - "create new cultures."
• Describe the cultural change that occurred as society shifted from Roman to Christian philosophy.
• Consider the quote, "The very struggle to become a Christian... seemed to be a struggle against astrology." How might the reaction to science by some Church leaders have been influenced by their reaction to the materialistic thinking of the Greeks and Romans?

WEEK 2

⬚ 30m General Science: Grade 7 - Lesson 51

The Material Versus the Spiritual

▯ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.22 p.193-197 "In the meantime," - "barbarians had won."
• Contrast Cyril's and Augustine's reactions to the events and thinking of their time. Why do you think they reacted so differently?

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

• IMPORTANT DATES
Augustine (354 A.D.-430 A.D.)

WEEK 2

⬚ 30m General Science: Grade 7 - Lesson 52

Thinking about Integrity

▯ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.22 p.198-199 "Caius Julius Solinus" - "and Essence"
• What was the Polyhistor?

WEEK 3

⬚ 30m General Science: Grade 7 - Lesson 53

Rejection of the Greeks

▯ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.23 p.200-205 "In 1529," - "improved on Ptolemy."
• What do you think about the term "Dark Ages"? Is it an accurate representation of the period or not?
• Some writers have suggested that society needed a period of penitence after Greek and Roman rule. Discuss this idea. Why would this be needed? What would such penitence look like?
• Charlotte Mason advised (Vol 3 p.49), 'we must bear in mind that truth behaves like a country

• IMPORTANT DATES
Middle Ages (476 A.D.-1000 A.D.)

Term 3

gate allowed to 'swing to' after a push. Now it swings a long way to this side and now a long way to that, and at last, after shorter and shorter oscillations, the latch settles.' If she is right, what do you expect will happen next in the Story of Science?

WEEK 3

⬚ **30m General Science: Grade 7 - Lesson 54**
The Gate Swings

Materials: Aristotle Leads the Way

➔ RECAP
What did you read last time?

➔ READ, NARRATE, & DISCUSS
Ch.24 p.206-209 "Pope Sylvester II" - "not encouraged."
• What's the 'big idea' or 'main purpose' of this chapter?
• How do the ideas connect back to previous chapters? At the same time, how does it add to the story?

➔ SUPPLEMENTAL
∞ Video Link: How to Use an Astrolabe (4:56)
∞ Video Link: Beauty of an Astrolabe (9:09)

WEEK 4

⬚ **30m General Science: Grade 7 - Lesson 55**
The Gate Swings

Materials: Aristotle Leads the Way

➔ RECAP
What did you read last time?

➔ READ, NARRATE, & DISCUSS
Ch.24 p.210-217 "Around 1110," - "called samurai."
• Discuss the role of learning and education in society. Is learning a basic human need? What role does curiosity hold? Should all ideas be completely free and available for everyone?

• IMPORTANT DATES
Adelard translates Euclid into Latin (c.1100 A.D.)

WEEK 4

⬚ **30m General Science: Grade 7 - Lesson 56**
Mathematical Revolution

Materials: Aristotle Leads the Way

➔ RECAP
What did you read last time?

➔ READ, NARRATE, & DISCUSS
Ch.25 p.218-221 "Do you want" - "handle its basics."
• Explain: "The Indian thinkers took three ideas that had been used elsewhere and made a powerful connection between them."

➔ SUPPLEMENTAL
∞ Video Link: Why Can't You Divide by Zero? (4:51)

• IMPORTANT DATES
Brahmagupta (c.598 A.D.-c.660 A.D.)

WEEK 5

⬚ **30m General Science: Grade 7 - Lesson 57**
Mathematical Revolution

Materials: Aristotle Leads the Way

PREP: Read Teacher Tip

➔ RECAP
What did you read last time?

➔ READ, NARRATE, & DISCUSS
Ch.25 p.221-225 "Someone had to" - "you do today."
• What's the 'big idea' or 'main purpose' of this chapter?
• How do the ideas connect back to previous chapters? At the same time, how does it move the story forward?

★ TEACHER TIP
Are your learner(s) demonstrating more familiarity/interest in math or its relationship with science? More interest/ease in discussing the complex relationship between science and society? If not, consider observing and discussing together more often, exploring extra helpings, or taking a field trip to a science museum. Reach out through Contact Us, if you need help.

• IMPORTANT DATES
House of Wisdom (est. 830 A.D.), Islamic Golden Age (800 A.D.-1258 A.D.), Leonardo Pisano [Fibonacci] (c.1170 A.D.-c.1250 A.D.)

WEEK 5

⬚ **30m General Science: Grade 7 - Lesson 58**
Fibonacci's Numbers

Materials: Aristotle Leads the Way

➔ RECAP
What did you read last time?

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

Term 3

→ READ, NARRATE, & DISCUSS
Ch.25 p.226-227 "Fibonacci" - "not always exact."
• What are Fibonacci's numbers conceptually?

WEEK 6

⬚ 30m General Science: Grade 7 - Lesson 59
Aquinas and the Truce

Materials: Aristotle Leads the Way

PREP: Read Teacher Tip

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.26 p.228-232 "In 1225," - "job of it."
• Tell about Aquinas' life.
• Contrast learning in a monastery with learning in a university, according to the author. What scientific question is in the air at the university at this time?

★ TEACHER TIP
Remember to engage students on their reading through natural discussion. If you need help, use the bulleted prompts.

• IMPORTANT DATES
Thomas Aquinas (1225-1274 A.D.), Roger Bacon (c.1214-1292 A.D.)

WEEK 6

⬚ 30m General Science: Grade 7 - Lesson 60
Aquinas and the Truce

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.26 p.233-235 "In Paris," - "above it all."
• We have now seen men of faith both promote scientific thinking and caution against scientific thinking. Is faith at the root of this conflict? If so, explain. If not, what do you think is?

→ SUPPLEMENTAL
The Triumph of Saint Thomas Aquinas was the subject of more than one artist (the author mentions one version on p.235). Consider two of them at the links. What do you notice, and what does this have to do with his message about truth and learning?
∞ Image Link: Triumph by Gozzoli
∞ Image Link: Triumph by da Firenze

WEEK 7

⬚ 30m General Science: Grade 7 - Lesson 61
Imagination Predicts Innovation

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.26 p.236-237 "Like many" - "glittering jewels."
• Describe some of the future inventions predicted by Roger Bacon. What kind of thinking allows scientists (or science fiction writers) to make such predictions? Are they predictions or are they self-fulfilling?

WEEK 7

⬚ 30m General Science: Grade 7 - Lesson 62
The Power of Books

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.27 p.238-242 "Johannes Gutenberg" - "work of a genius."
• What's the 'big idea' or 'main purpose' of this chapter?
• How do the ideas connect back to previous chapters? What do you think will happen next?

• IMPORTANT DATES
Gutenberg prints first book on printing press (1445 A.D.)

WEEK 8

⬚ 30m General Science: Grade 7 - Lesson 63
The Power of Books

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.27 p.242-247 "We don't know" - "we be now?"
• How can people learn to differentiate between right and wrong, correct and incorrect ideas? Is it

• NATURE NOTEBOOK
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Term 3

always bad to believe an idea even if it turns out to be incorrect? When is it okay and when is it not okay? How do we know?
• What was the underlying source of tension between some scientific and religious leaders during the time period of this book? (Look back at Ch.5, 26, 27, if you need examples.)

WEEK 8

⬚ 30m General Science: Grade 7 - Lesson 64

What Magellan's Journey Meant

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.28 p.248-252 "On September 4," - "about the Americas."
• What role does Magellan's accomplishment have in the story of science?
• Was Magellan a scientist? Why or why not?

WEEK 9

⬚ 30m General Science: Grade 7 - Lesson 65

What Magellan's Journey Meant

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.28 p.253-257 "A seasoned soldier" - "changed everything."
• "Magellan changed everything." Explain.

• **IMPORTANT DATES**
Magellan sails around the world (1519-1522 A.D.)

WEEK 9

⬚ 30m General Science: Grade 7 - Lesson 66

Imagining a Journey

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.29 p.258-265 "Far, far from" - "we were coming."
• What's the 'big idea' or 'main purpose' of this chapter?
• How do the ideas connect back to previous chapters? At the same time, how does it lead us forward?

★ TEACHER NOTE
A short science fiction story is presented in which the timescale of Earth's existence is important on p.261: "Thuleans send" - "of particular interest." Teachers might choose to allow it to pass by the way or to discuss with students.

WEEK 10

⬚ 30m General Science: Grade 7 - Lesson 67

It's About Uncertainty

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.30 p.266-269 "There is something" - "the last word."
• "Science is not about certainty; it is about uncertainty." What do you think the author means by this? What does this mean for you?
• Why does the author claim that good scientists are humble? What other habits do you think they have?

★ TEACHER NOTE
The age of the Earth is mentioned on p.268: "Those two theories" - "fixed, permanent stage." Teachers might choose to allow it to pass by the way, to discuss with students, or to consider alternative theories they wish to share.

WEEK 10

⬚ 30m General Science: Grade 7 - Lesson 68

Catch Up

Materials: Aristotle Leads the Way

→ CATCH UP
Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

WEEK 11

⬚ 30m General Science: Grade 7 - Lesson 69

Catch Up

Materials: Aristotle Leads the Way

→ CATCH UP
Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

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General Science: Grade 7

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Term 3

WEEK 11	⬚ **30m General Science: Grade 7 - Lesson 70** *Catch Up* ▯ Materials: Aristotle Leads the Way PREP: Read Teacher Tip **➔ CATCH UP** Use this time to catch up on lessons in this or another course, or explore Extra Helpings.	★ TEACHER TIP When student(s) take exams next week, evaluate what they recall, but more importantly whether math, its place in science, or the relationship between science and society are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc? Reach out through Contact Us, if you need help.
WEEK 12	⬚ **30m General Science: Grade 7 - Lesson 71** *Exam Week* **➔ EXAMS** Answer question(s) related to the course. Questions will come from: Aristotle Leads the Way	
WEEK 12	⬚ **30m General Science: Grade 7 - Lesson 72** *Exam Week* **➔ EXAMS** Answer question(s) related to the course. Questions will come from: Aristotle Leads the Way	

General Science: Grade 7

Examination

Term 1

· Aristotle Leads the Way: Explain at least two examples of how careful record-keeping and/or measurement advanced science in the ancient world.

· Aristotle Leads the Way: What were some obstacles that made science difficult for early scientists, and what urged them forward?

· Explain how you did one laboratory investigation this term and what you learned from it.

Term 2

· Aristotle Leads the Way: We are told that "Plutarch said that Archimedes believed engineering was 'sordid and ignoble, [as is] every sort of art that lends itself to mere use and profit.'" Why then did he design so many useful devices? Describe 2 of them.

· Aristotle Leads the Way: Explain why Alexandria was different from other cities of its time. How did it help to create a new age of technological and intellectual advancement?

· Explain how you did one laboratory investigation this term and what you learned from it.

Term 3

· Aristotle Leads the Way: Explain how developments in mathematics changed the way scientists learned about their world.

· Aristotle Leads the Way: Were the Dark Ages really intellectually dark? Give reasons to support your conclusion.

· Explain how you did one laboratory investigation this term and what you learned from it.